C-DOT control science

Phase 2.1 → Phase 3.2

Every experiment · every gate · every boundary

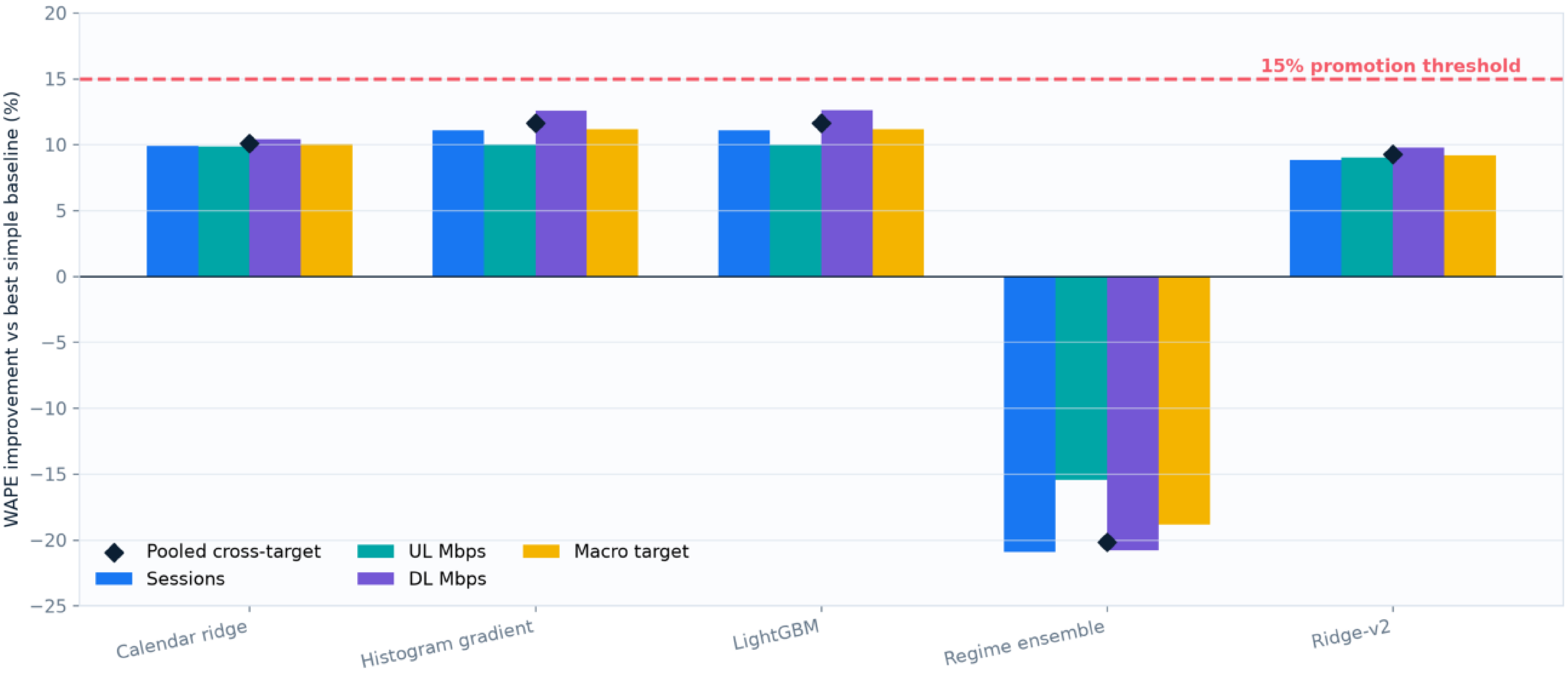
Evidence scale

588 paired one-day runs + 125 survival trials

Outcome: no candidate passed every gate · Static remains production default

Forecast gains are real—but below the promotion bar

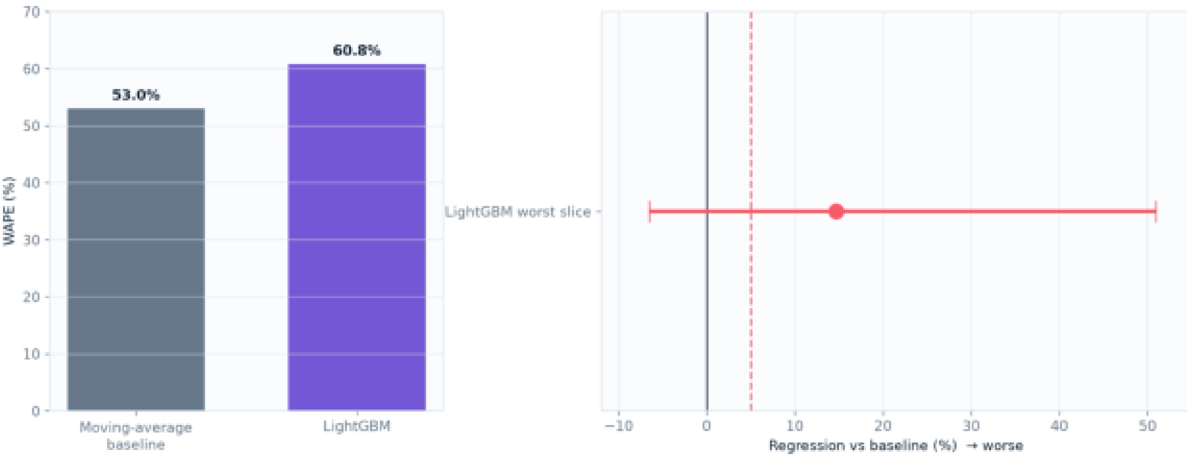
Target-separated WAPE prevents unlike units from being hidden inside the pooled headline



Seed 46003 generated + sealed · never evaluated or selected point estimate 14.7%

The worst LightGBM slice is visible—and statistically fragile

Detected-surge DL load at 30 minutes: n=9 observations across 8 groups



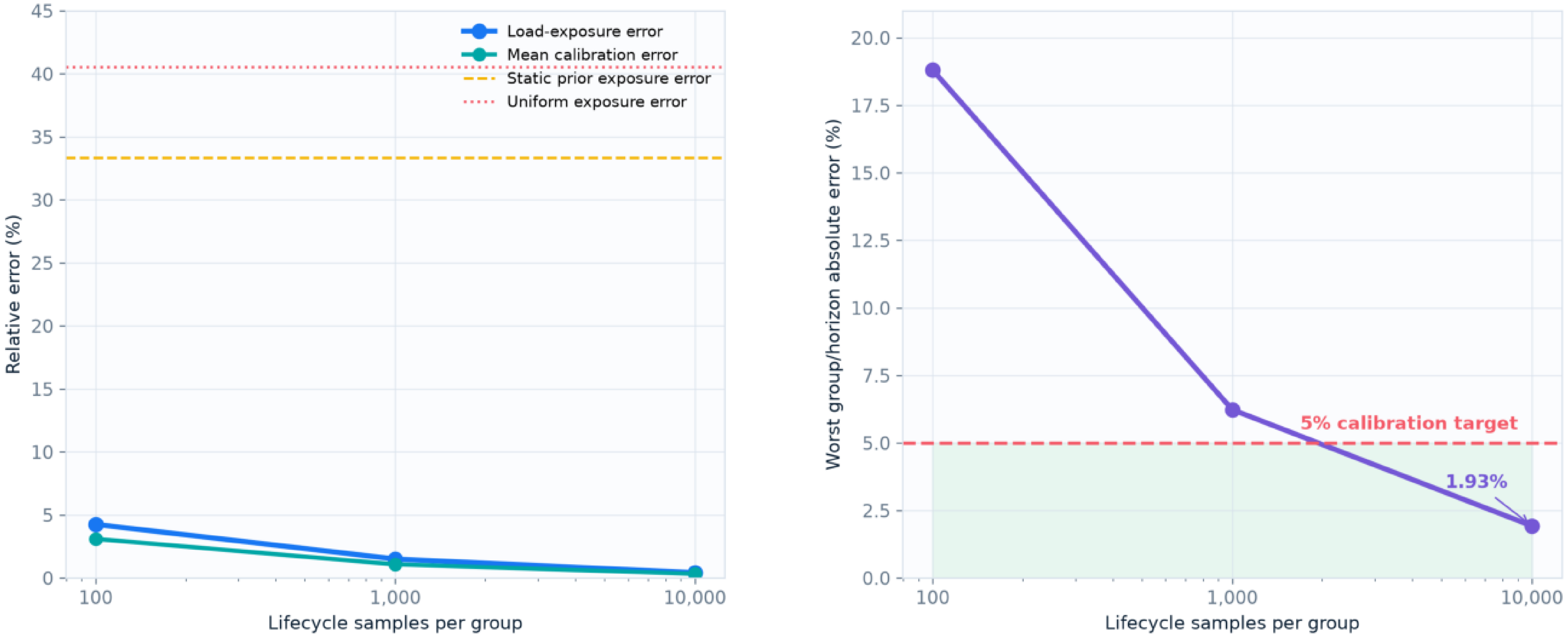
Interpretation: keep the regression visible, but do not present nine observations as a stable population estimate.

Seed 46003 generated + sealed · never evaluated or selected

95% group-cluster CI [-6.5%, 50.9%]

Kaplan–Meier mechanics converge on synthetic lifecycle telemetry

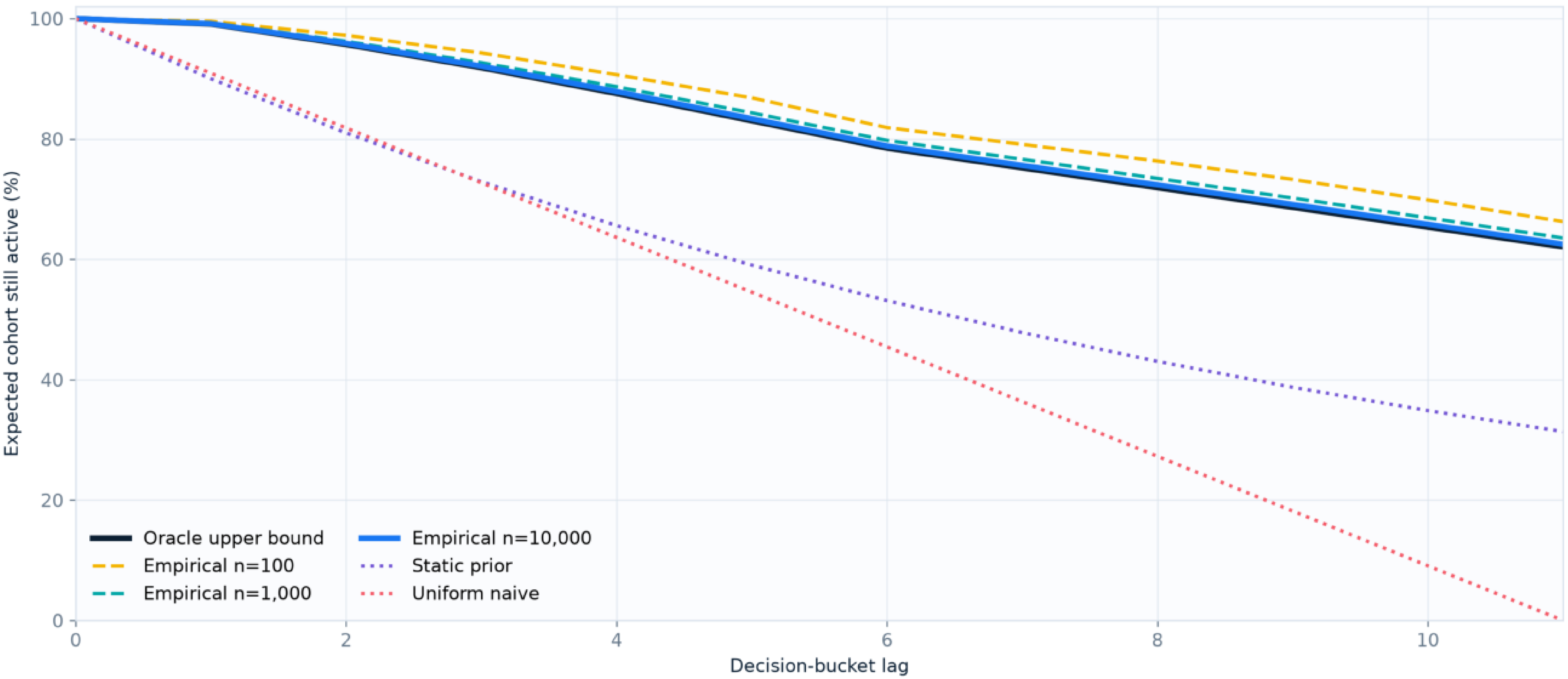
Synthetic censored records test estimation mechanics—not real-world distribution independence



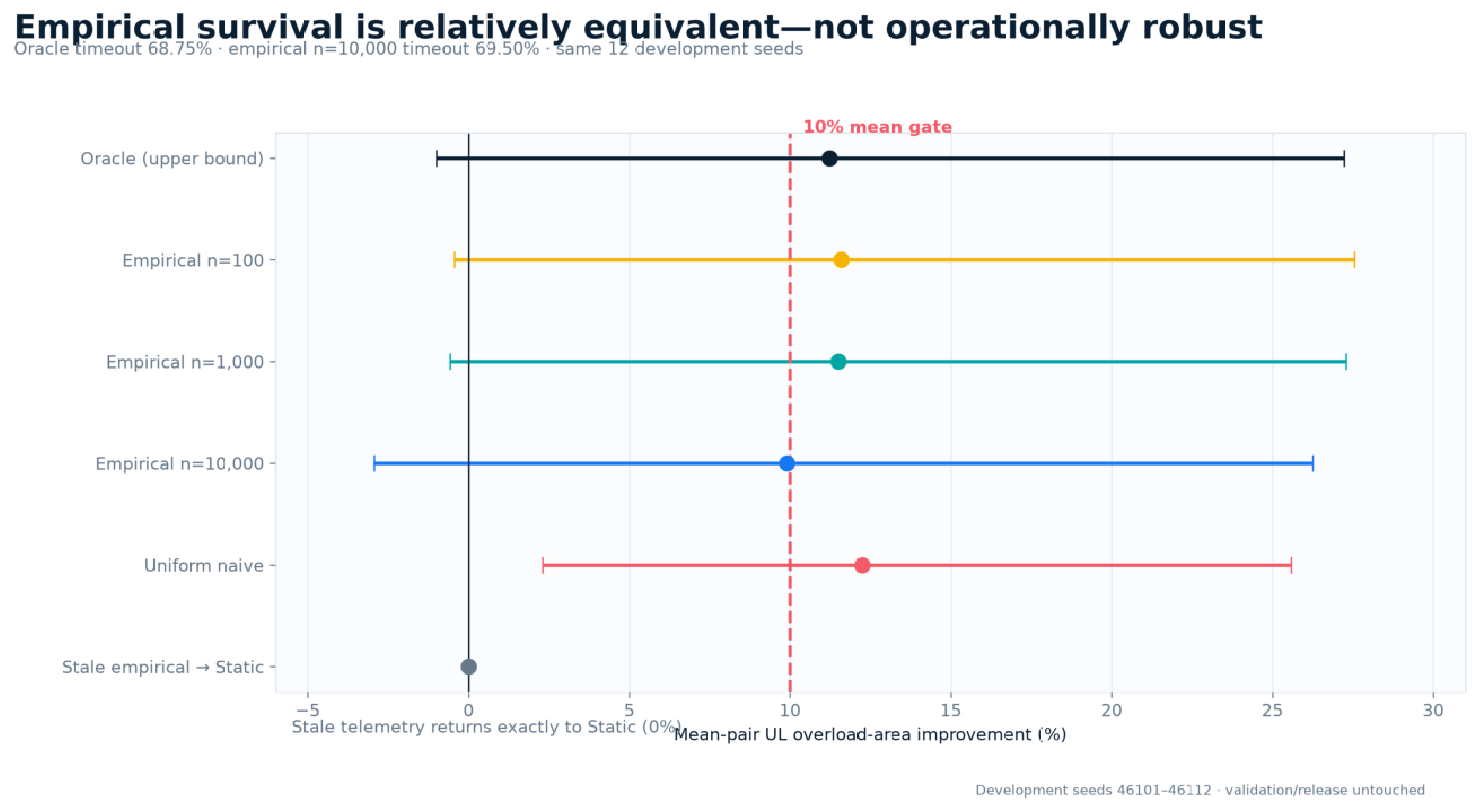
Development seeds 46101–46112 · validation/release untouched

What the MPC actually consumes: expected cohort survival

Average across 96 traffic groups; empirical n=10,000 closely tracks the non-deployable oracle

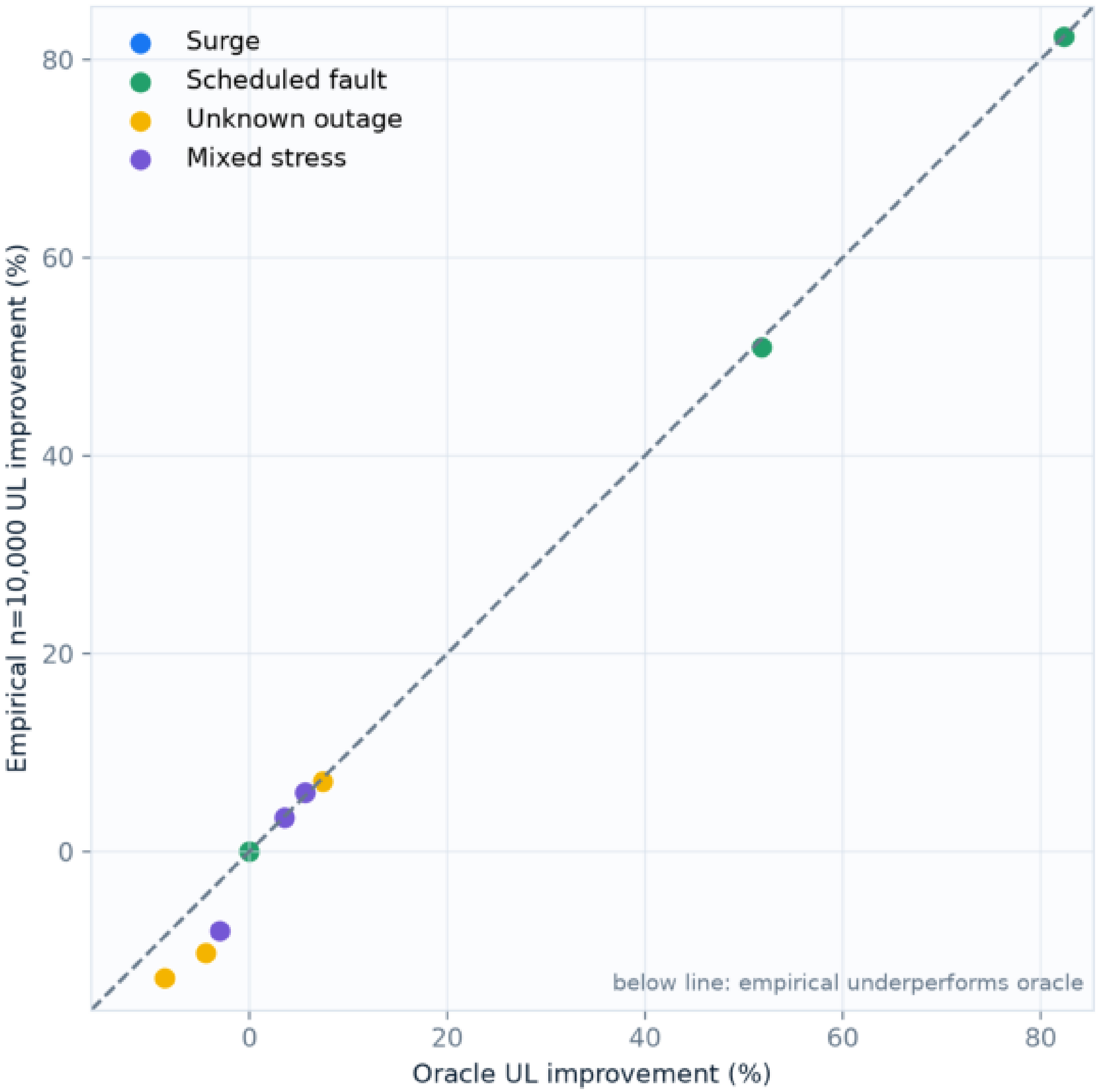


Development seeds 46101–46112 · validation/release untouched


Empirical survival is relatively equivalent—not operationally robust
Oracle timeout 68.75% · empirical n=10,000 timeout 69.50% · same 12 development seeds
10% mean gate
Oracle (upper bound)
Empirical n=100
Empirical n=1,000
Empirical n=10,000
Uniform naive
Stale empirical → Static
−5
0
5
10
15
20
25
30
Stale telemetry returns exactly to Static (0%)
Mean-pair UL overload-area improvement (%)
Development seeds 46101–46112 · validation/release untouched

Imperfect survival follows oracle behavior pair by pair

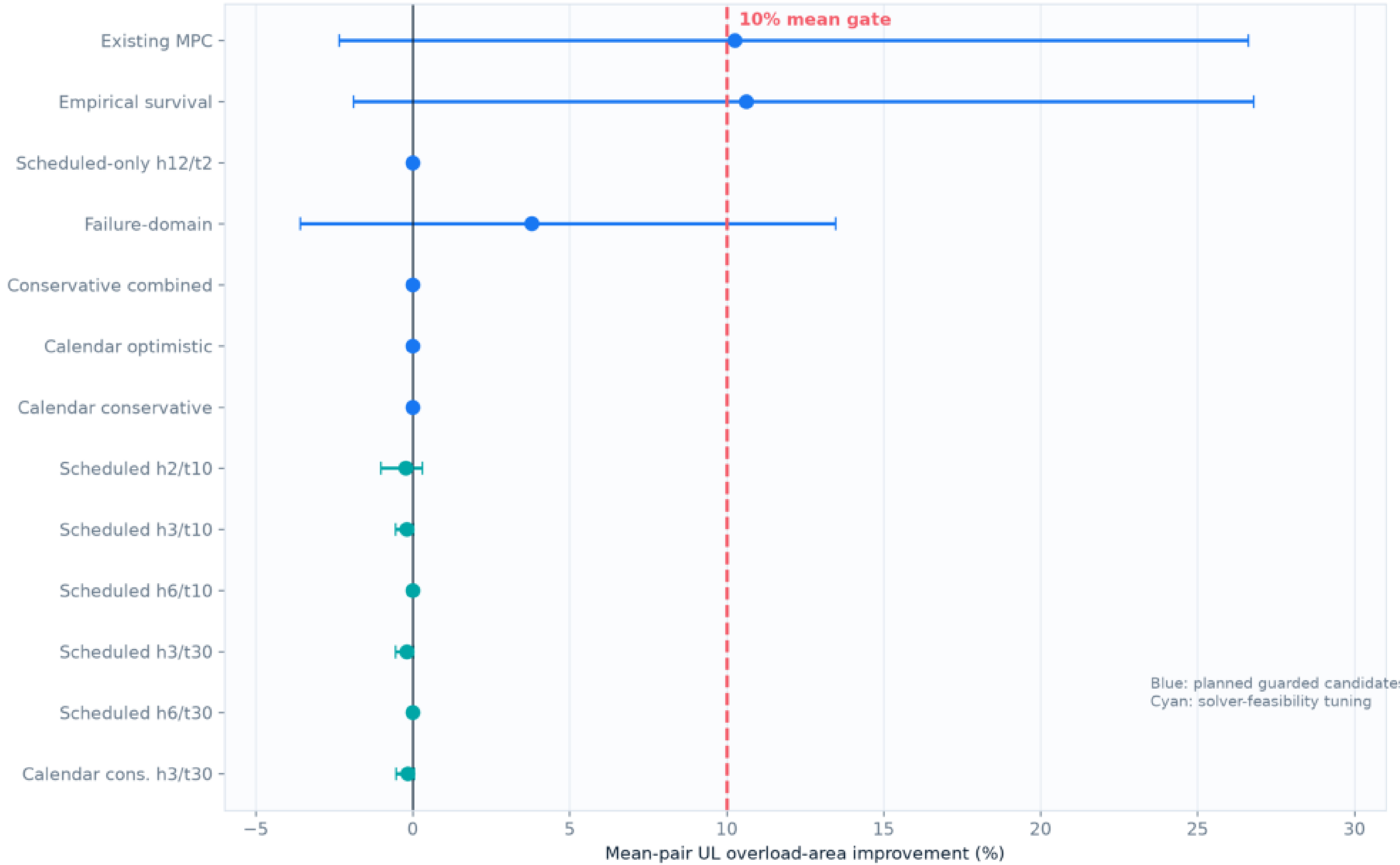
Empirical n=10,000 versus oracle upper bound; 12 identical development pairs



Development seeds 46101–46112 · validation/release untouched

All 13 guarded MPC candidates fail the frozen Day-5 bar

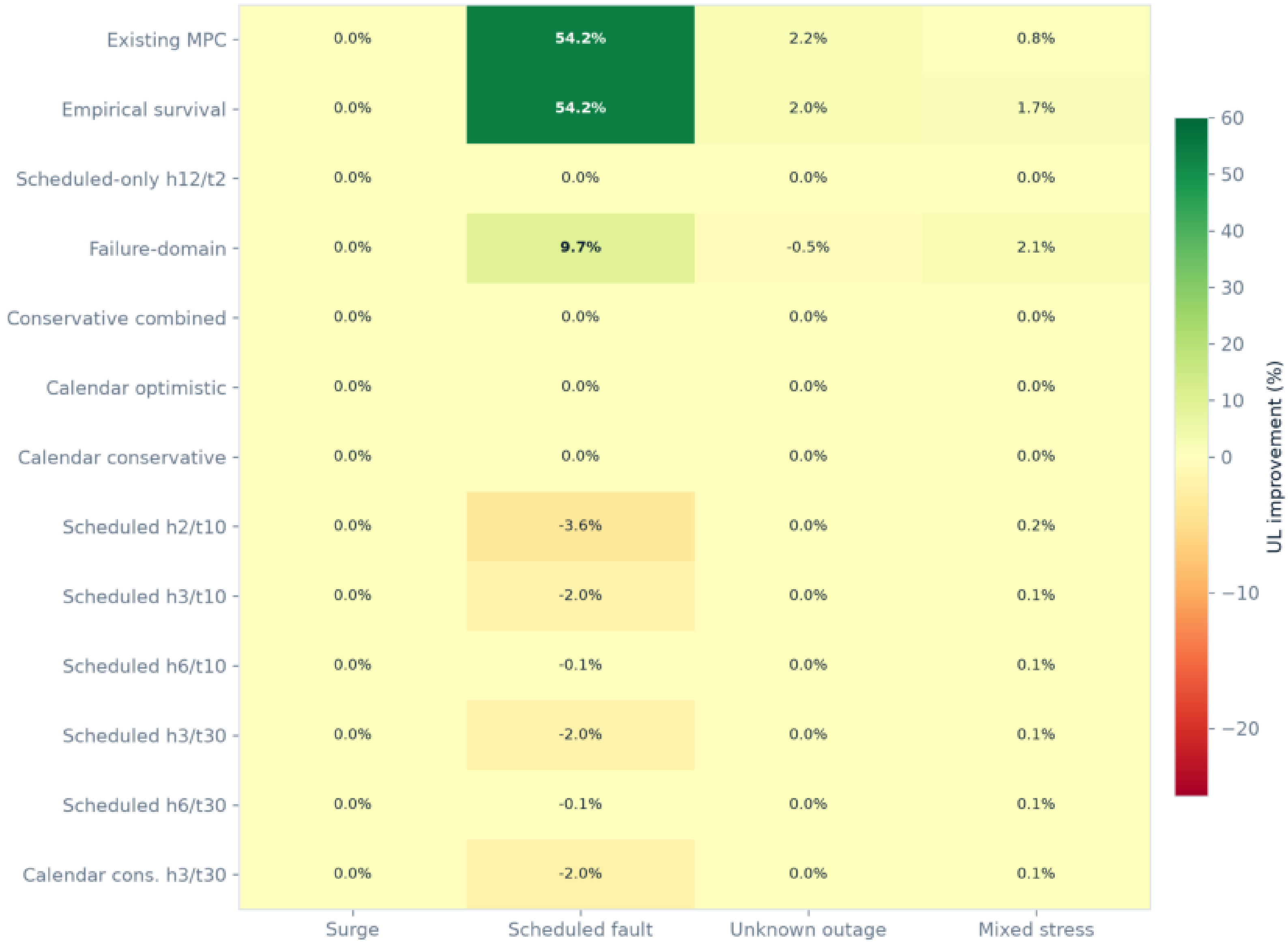
Forest plot shows paired mean UL improvement and bootstrap 95% interval



Development seeds 46101–46112 · validation/release untouched

Scenario view: broad MPC gains are concentrated and unstable

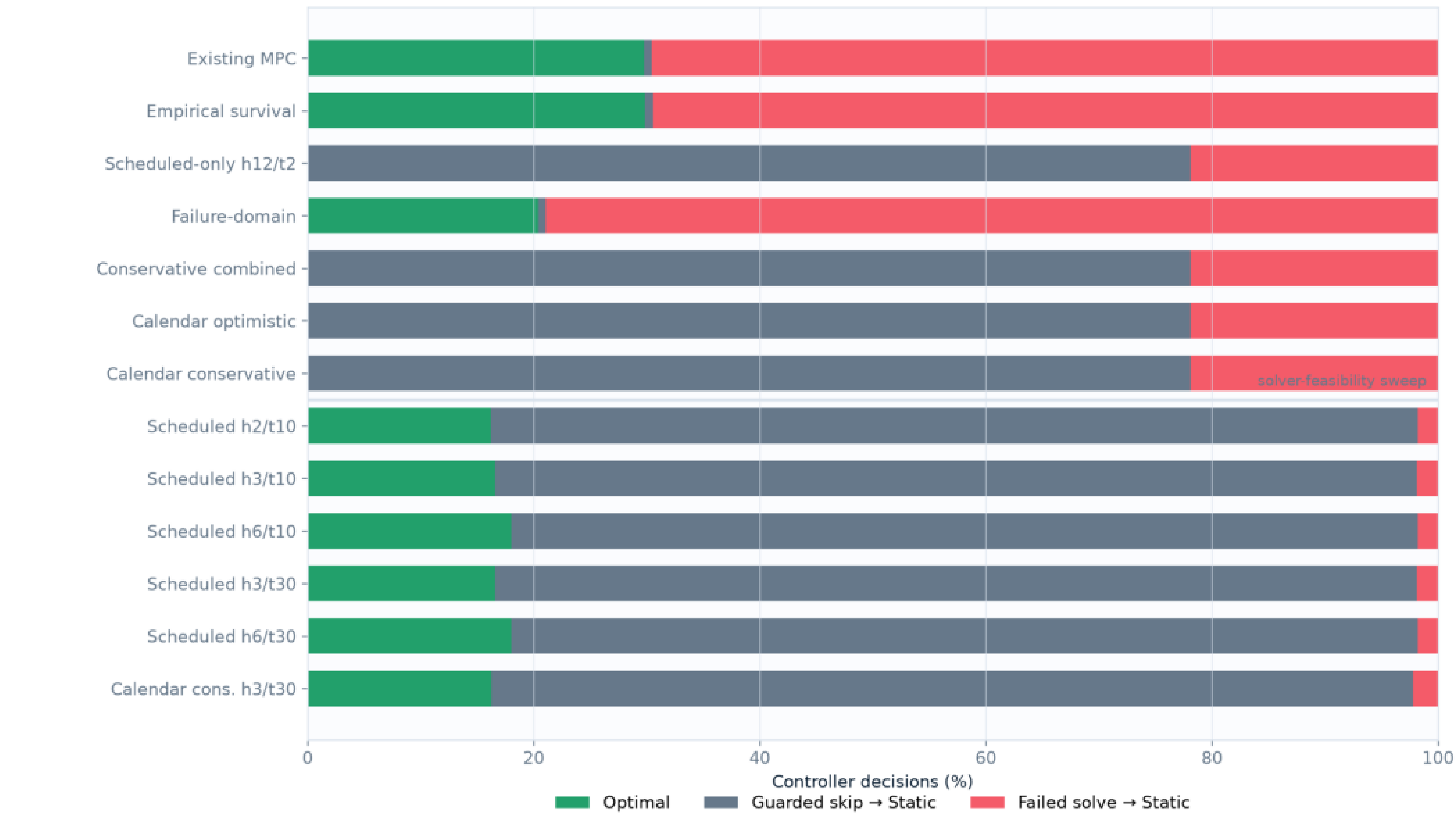
Aggregate UL overload-area improvement; three paired seeds per scenario



Development seeds 46101–46112 · validation/release untouched

Solver-feasibility tuning removed timeouts—not the performance gap

Share of controller decisions by solver status; failed = timeout, error, or infeasible



Development seeds 46101–46112 · validation/release untouched

Fail closed means every gate must pass

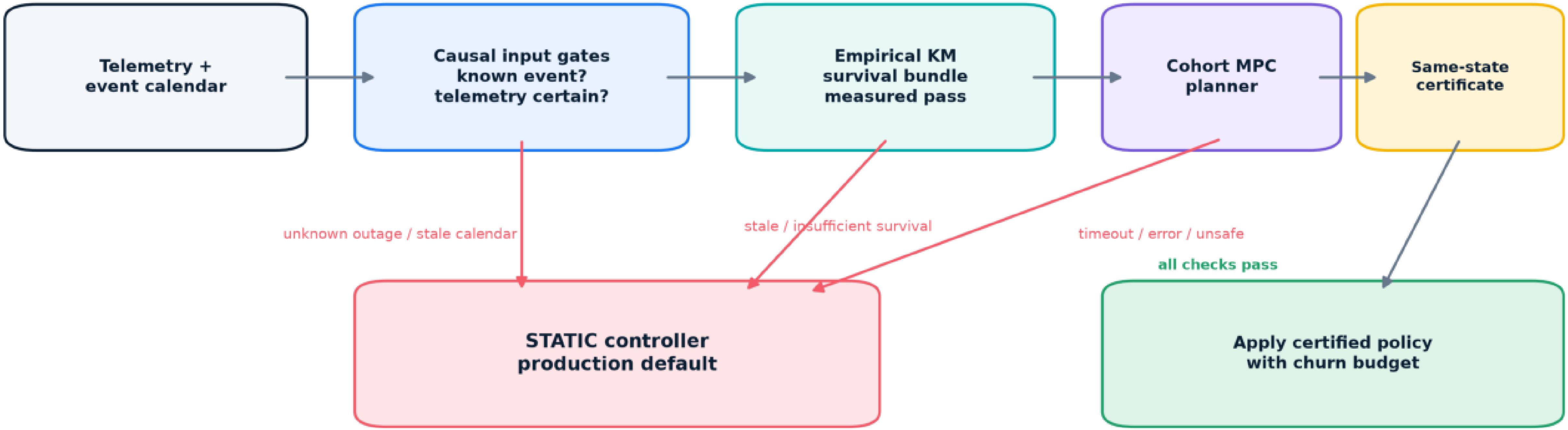
Green = pass, red = fail; no candidate clears all 11 frozen development gates

	Mean UL ≥10%	Bootstrap lower >0	Severity-weighted >0	Unknown/mixed ≥−2%	Worst pair >−10%	No DL/drop/session regression	No timeout/error	Unexpected fallback ≤1%	Skipped decisions ≤95%	Churn ≤0.05	Measured survival robust
Existing MPC	✓	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
Empirical survival	✓	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
Scheduled-only h12/t2	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Failure-domain	✗	✗	✓	✓	✗	✓	✗	✗	✓	✓	✓
Conservative combined	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Calendar optimistic	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Calendar conservative	✗	✗	✗	✓	✓	✓	✗	✗	✓	✓	✓
Scheduled h2/t10	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h3/t10	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h6/t10	✗	✗	✓	✓	✓	✓	✓	✗	✓	✓	✓
Scheduled h3/t30	✗	✗	✗	✓	✓	✗	✓	✗	✓	✓	✓
Scheduled h6/t30	✗	✗	✓	✓	✓	✓	✓	✗	✓	✓	✓
Calendar cons. h3/t30	✗	✗	✗	✓	✓	✗	✗	✗	✓	✓	✓

Development seeds 46101–46112 · validation/release untouched

Production architecture: MPC is an optional, guarded branch

Unknown outages, uncertain telemetry, stale survival, or failed certification return immediately to Static

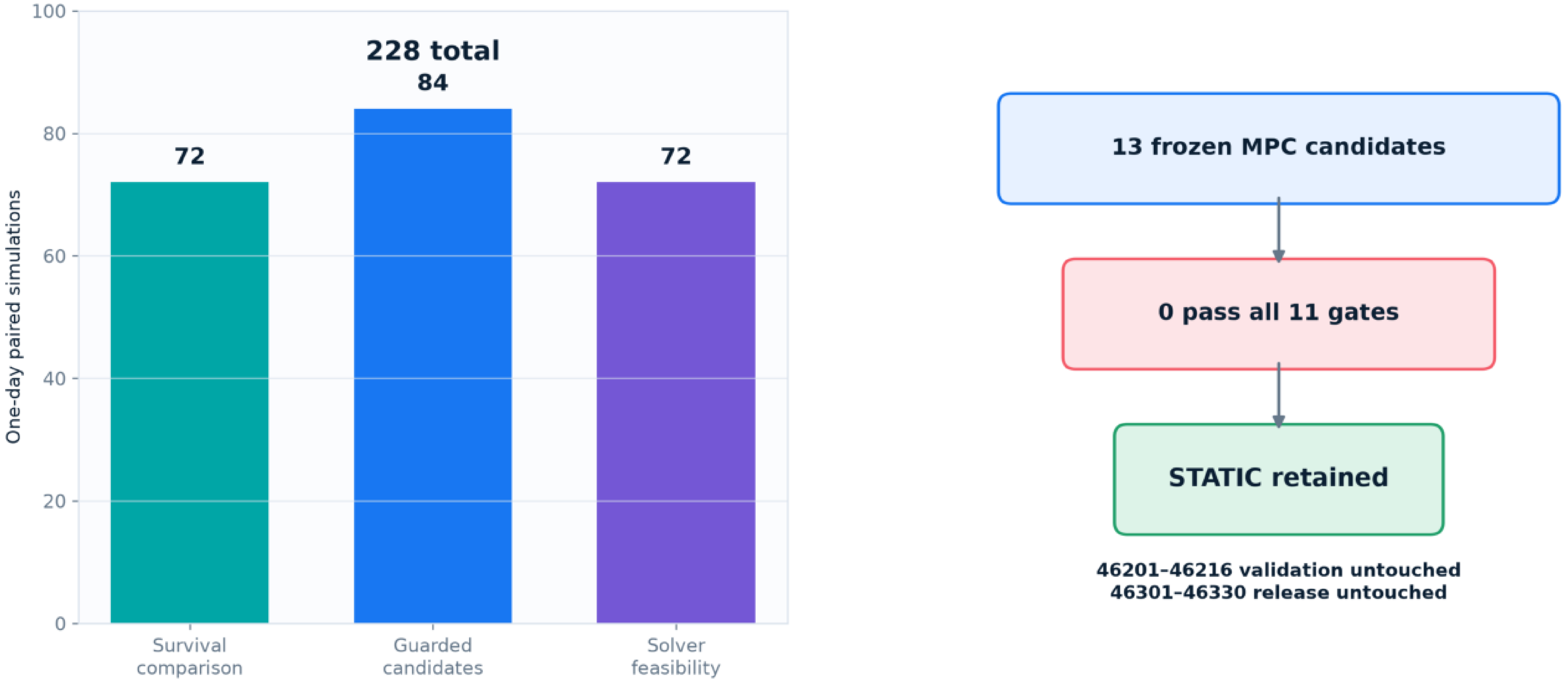


Measured result: the guardrail plumbing works; no tested MPC branch earned authority to replace Static.

Development seeds 46101–46112 · validation/release untouched

228 paired one-day runs support a disciplined stop decision

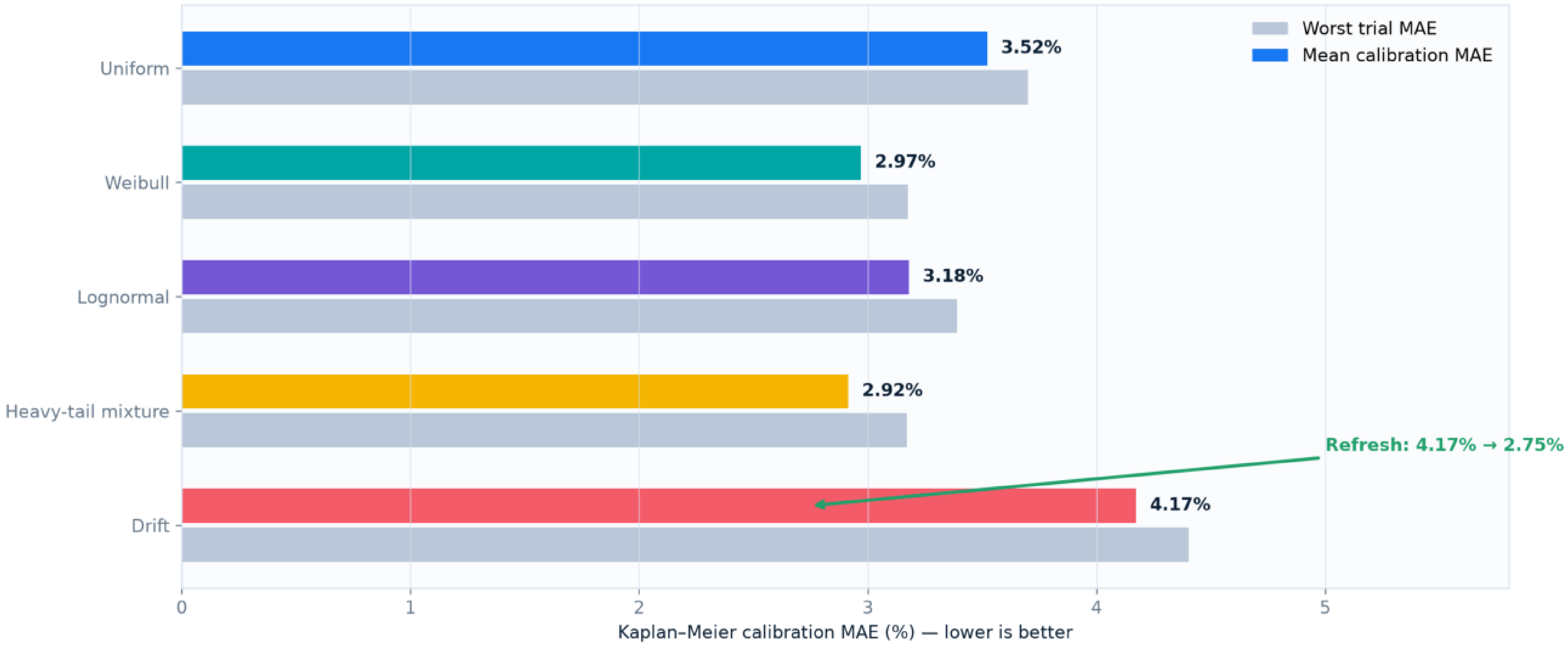
The campaign used development seeds repeatedly and preserved validation/release for a genuine winner



Development seeds 46101–46112 · validation/release untouched

Distribution-blind lifecycle survival works mechanically

125 auditor-hidden trials · fit sees only start, end, censor, class and timestamps



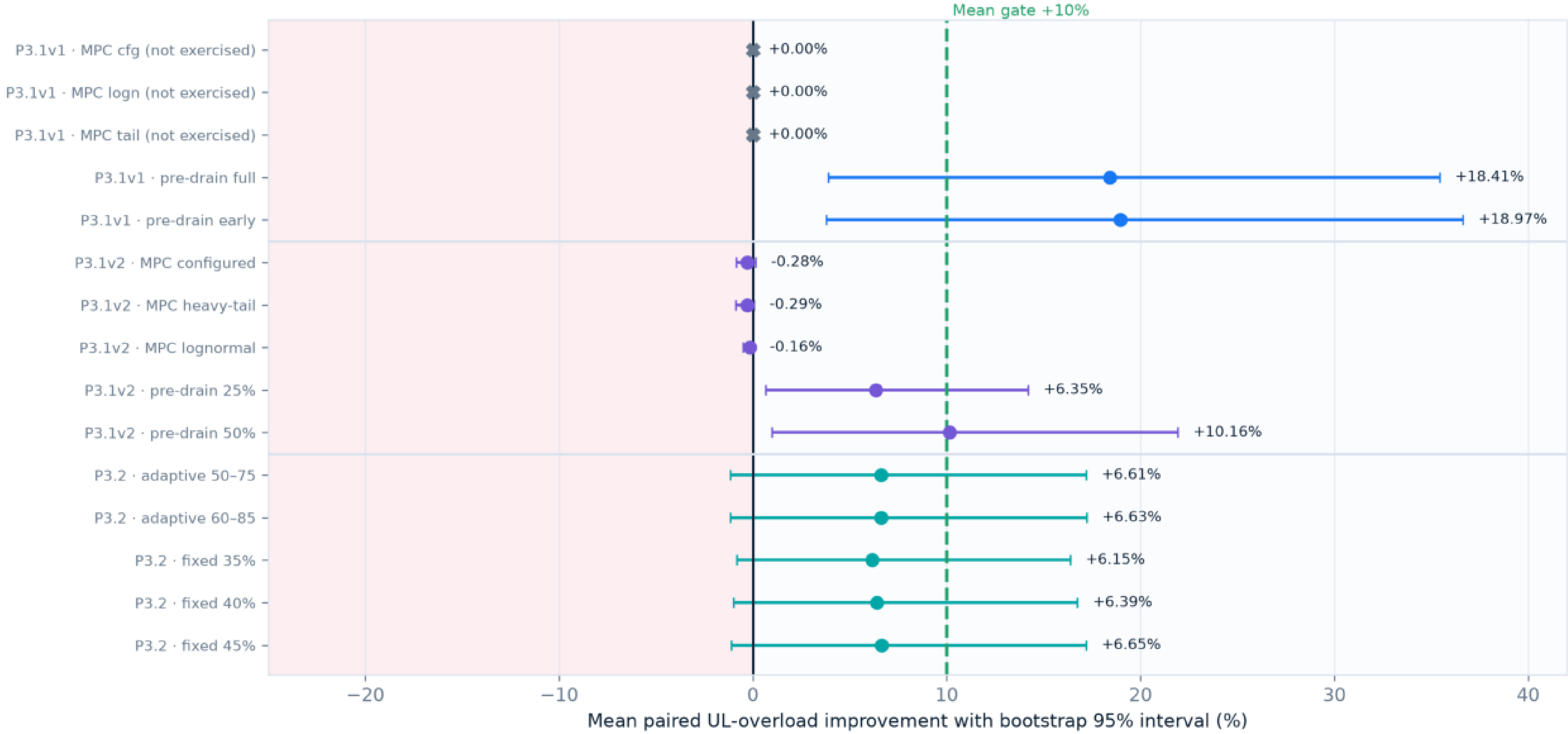
WORKED Staleness fail-closed 100% · pooling coverage 25%

BOUNDARY Simulated observations ≠ external real-world calibration

Fresh development seeds 46401–46472 · validation/release untouched

Every new development candidate: mean, uncertainty and mechanism status

360 paired one-day runs · confidence interval must exclude zero; mean must reach +10%

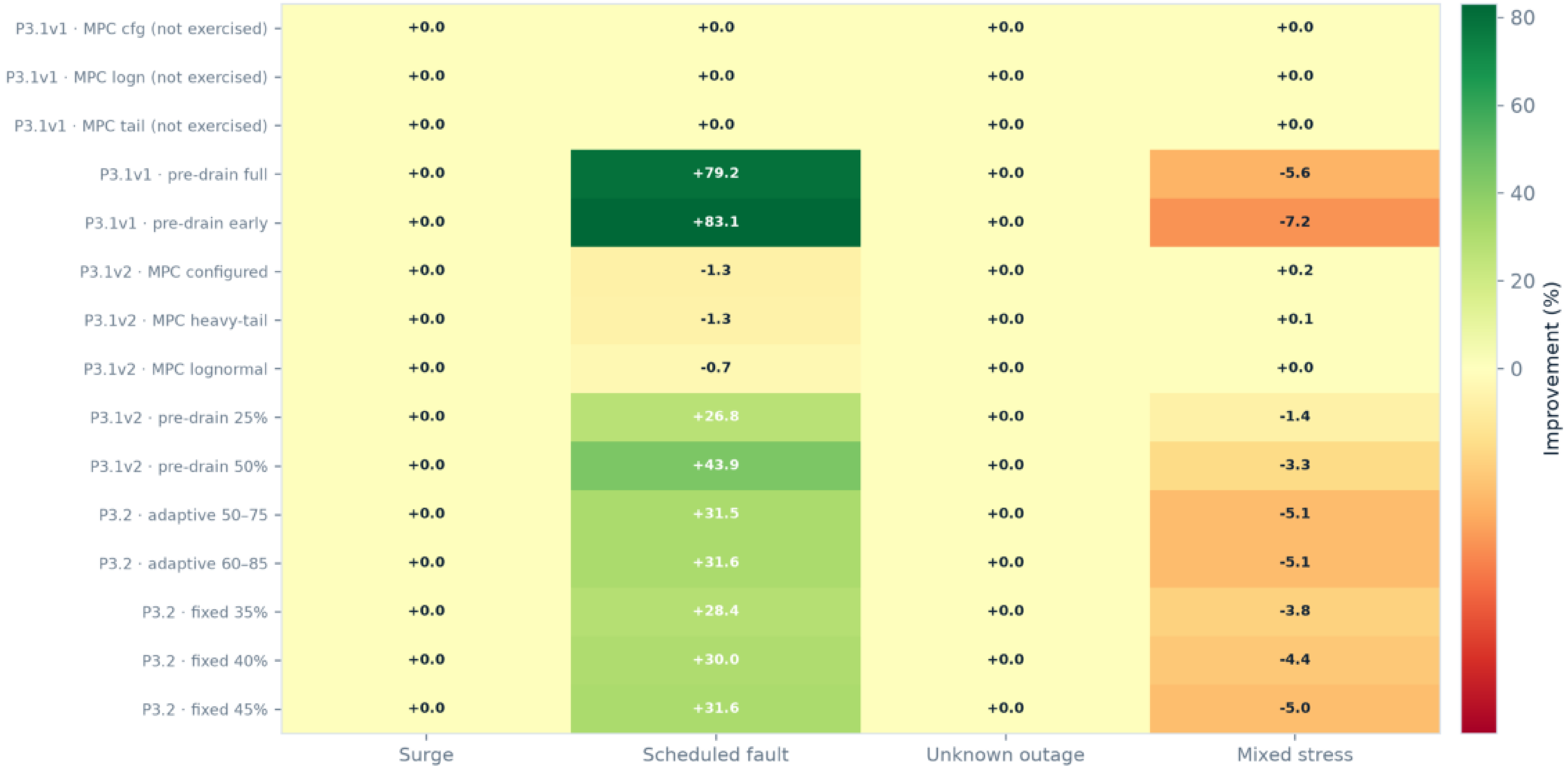


X = no proposed action: valid fail-closed, invalid mechanism exercise

Fresh development seeds 46401–46472 · validation/release untouched

Where gains came from—and where they failed

Scenario-stratified mean UL-overload improvement (%) · scheduled benefit did not generalize



WORKED: scheduled-fault pre-drain · DID NOT: robust transfer to mixed stress

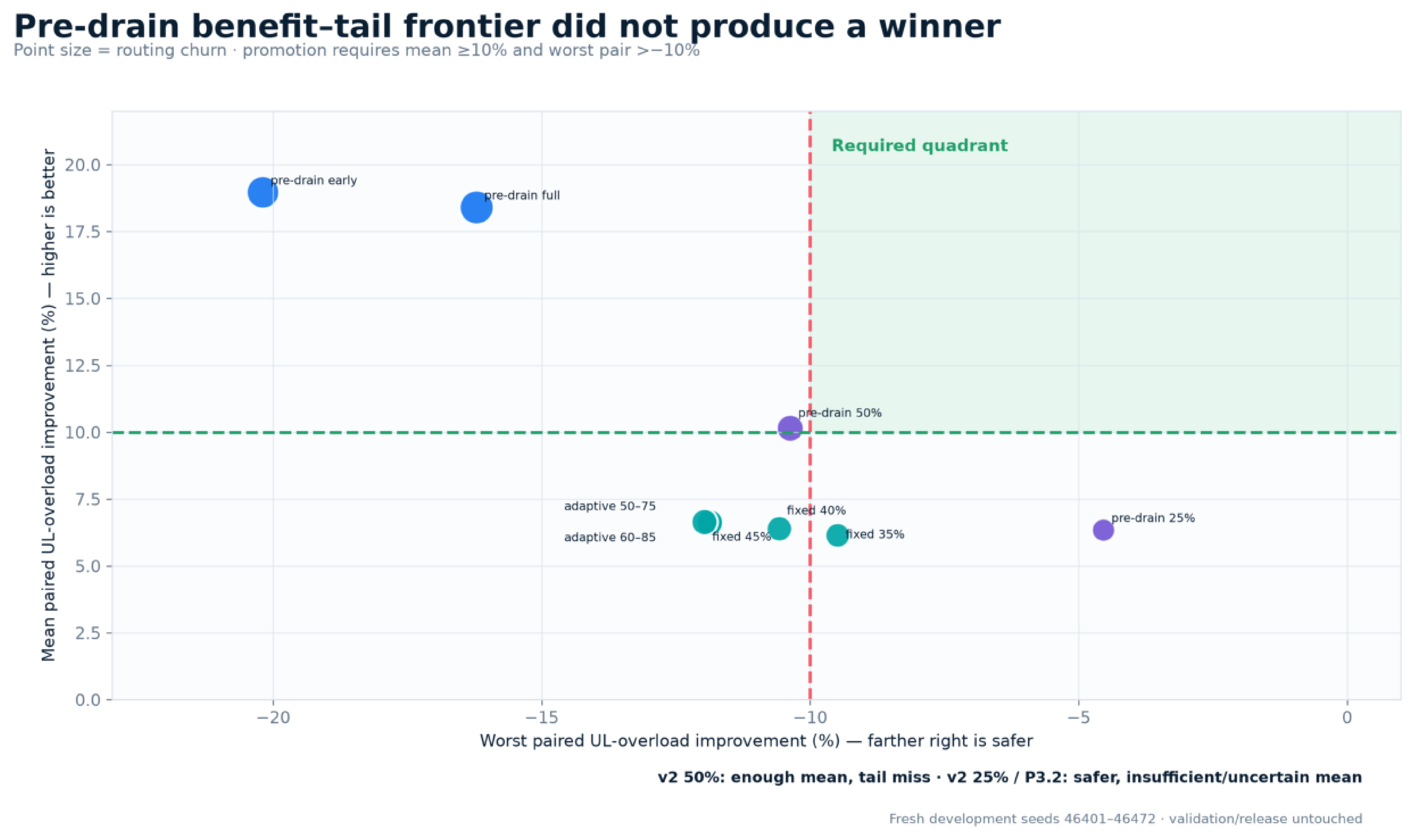
Fresh development seeds 46401–46472 · validation/release untouched

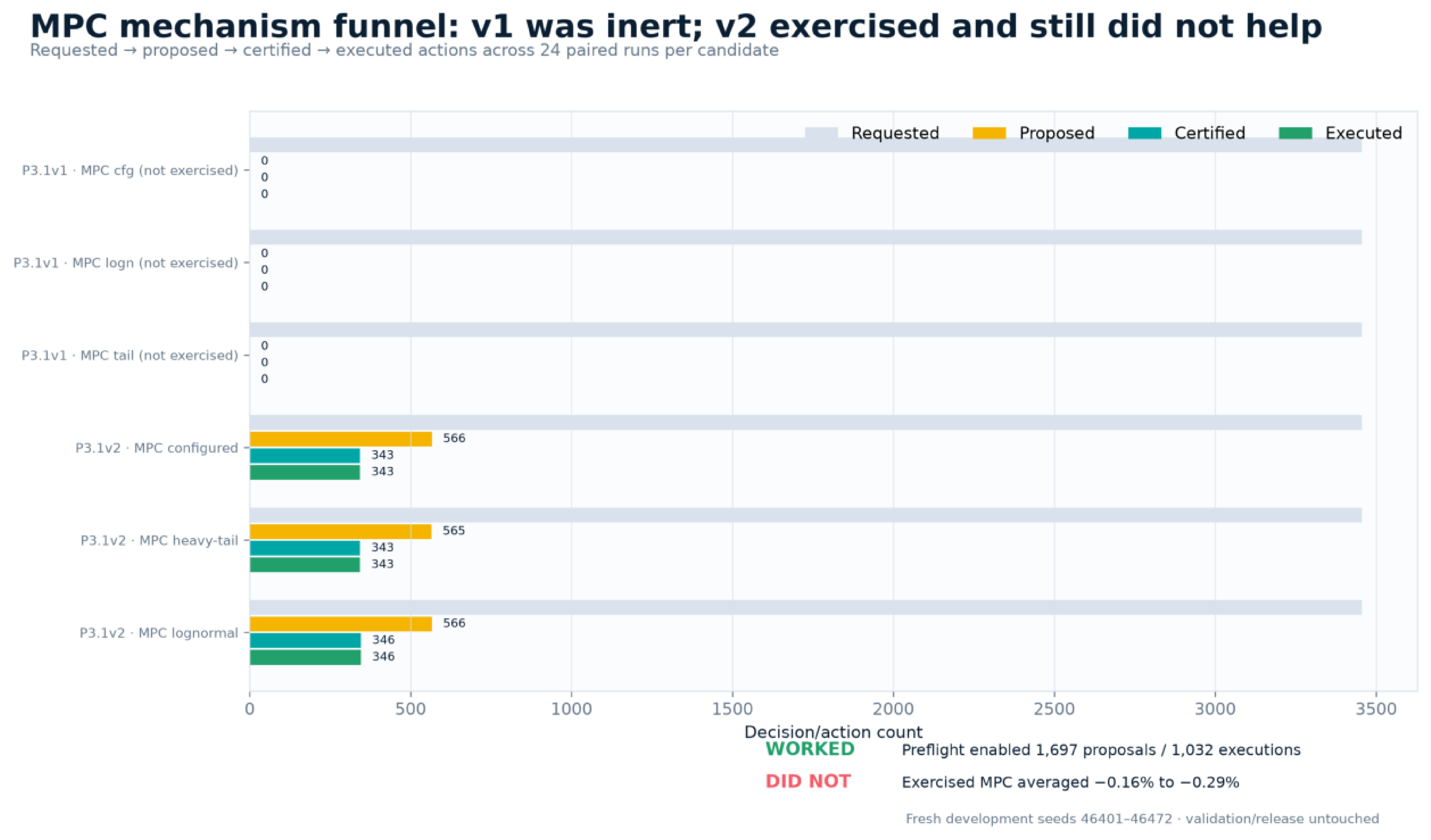
Complete promotion scorecard: no all-green row

Green = pass · red = fail · gray = gate not yet registered in that historical interface

	Mean UL ≥10%	Bootstrap lower >0	Severity-weighted >0	Unknown/mixed ≥−2%	Worst pair >−10%	No DL/drop/session regression	No timeout/error	Unexpected fallback ≤1%	Skipped decisions ≤95%	Churn ≤0.05	Measured survival robust	End-to-end deadline
P3.1v1 · MPC cfg (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—
P3.1v1 · MPC logn (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—
P3.1v1 · MPC tail (not exercised)	✗	✗	✗	✓	✓	✓	✓	✓	✗	✓	✗	—
P3.1v1 · pre-drain full	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	—
P3.1v1 · pre-drain early	✓	✓	✗	✓	✗	✓	✓	✓	✓	✗	✓	—
P3.1v2 · MPC configured	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—
P3.1v2 · MPC heavy-tail	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—
P3.1v2 · MPC lognormal	✗	✗	✓	✓	✓	✗	✓	✓	✓	✓	✗	—
P3.1v2 · pre-drain 25%	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—
P3.1v2 · pre-drain 50%	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	—
P3.2 · adaptive 50–75	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗
P3.2 · adaptive 60–85	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗
P3.2 · fixed 35%	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗
P3.2 · fixed 40%	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓
P3.2 · fixed 45%	✗	✗	✓	✓	✗	✓	✓	✓	✓	✓	✓	✗

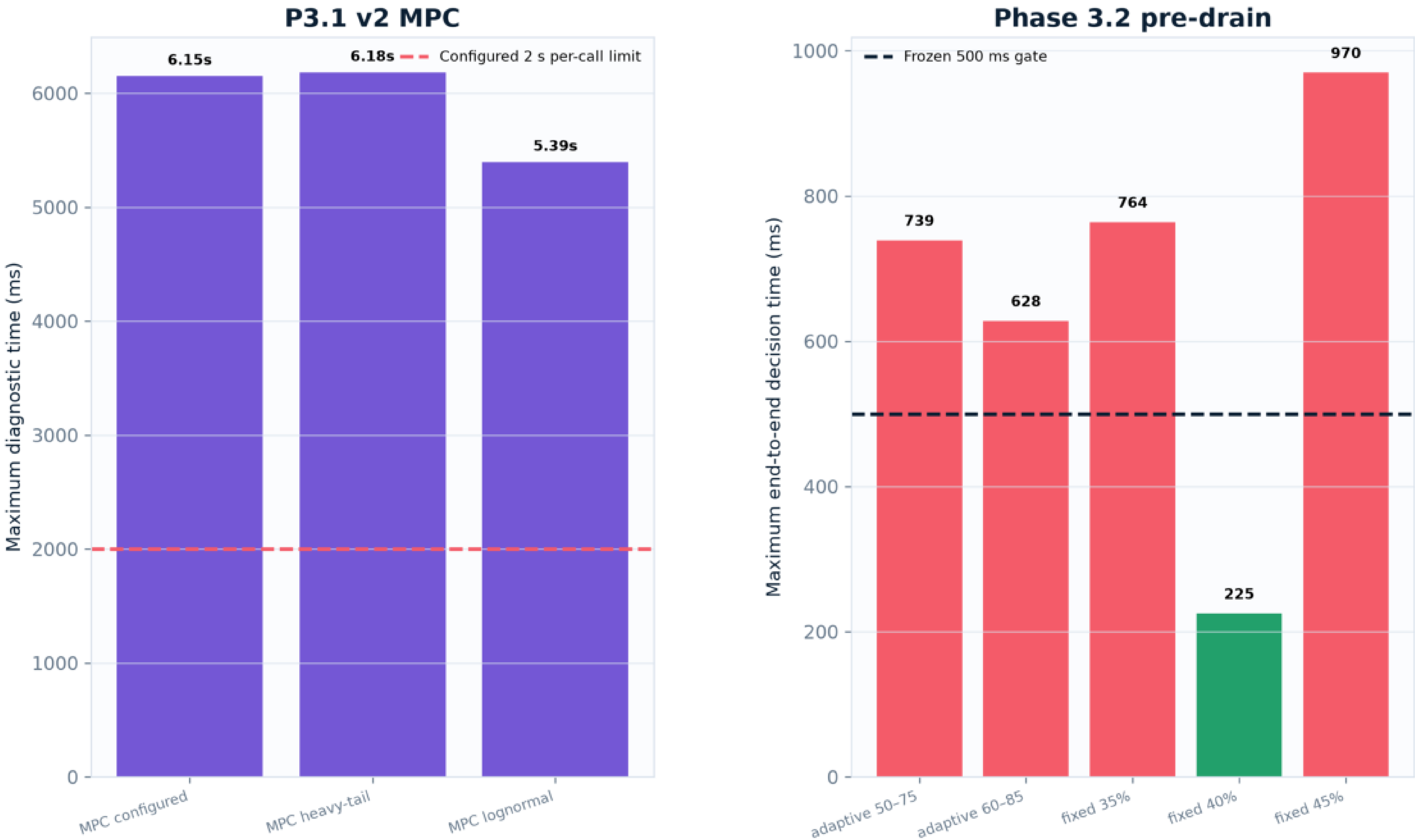
Fresh development seeds 46401–46472 · validation/release untouched


Pre-drain benefit–tail frontier did not produce a winner
Point size = routing churn · promotion requires mean ≥10% and worst pair >−10%
Required quadrant
pre-drain early
pre-drain full
pre-drain 50%
adaptive 50–75
adaptive 60–85
fixed 45%
fixed 40%
fixed 35%
pre-drain 25%
Mean paired UL-overload improvement (%) — higher is better
Worst paired UL-overload improvement (%) — farther right is safer
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−20
−15
−10
−5
0
v2 50%: enough mean, tail miss · v2 25% / P3.2: safer, insufficient/uncertain mean
Fresh development seeds 46401–46472 · validation/release untouched


MPC mechanism funnel: v1 was inert; v2 exercised and still did not help
Requested → proposed → certified → executed actions across 24 paired runs per candidate
Requested
Proposed
Certified
Executed
P3.1v1 · MPC cfg (not exercised)
P3.1v1 · MPC logn (not exercised)
P3.1v1 · MPC tail (not exercised)
P3.1v2 · MPC configured
P3.1v2 · MPC heavy-tail
P3.1v2 · MPC lognormal
0
0
0
0
0
0
0
0
0
566
343
343
565
343
343
566
346
346
0
500
1000
1500
2000
2500
3000
3500
Decision/action count
WORKED
Preflight enabled 1,697 proposals / 1,032 executions
DID NOT
Exercised MPC averaged −0.16% to −0.29%
Fresh development seeds 46401–46472 · validation/release untouched

Operational latency: solver status alone hid deadline risk

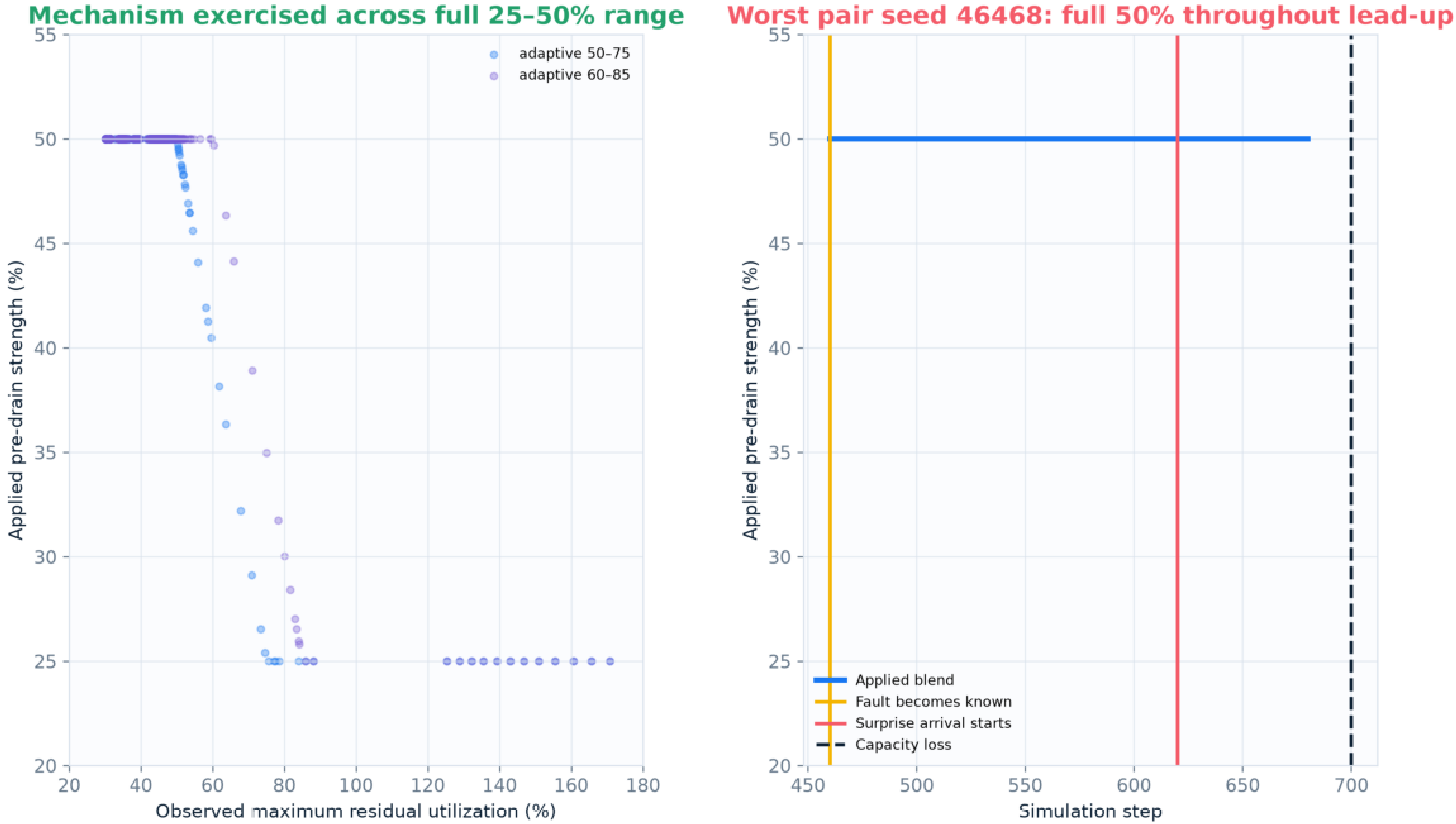
Left: MPC model+solve diagnostic versus 2 s per-call limit · right: full pre-drain decision versus 500 ms gate



Fresh development seeds 46401–46472 · validation/release untouched

Adaptive pre-drain worked as coded—but causal lag defeated the tail hypothesis

Applied blend responds to observed residual utilization; worst pair's surprise begins after commitment starts

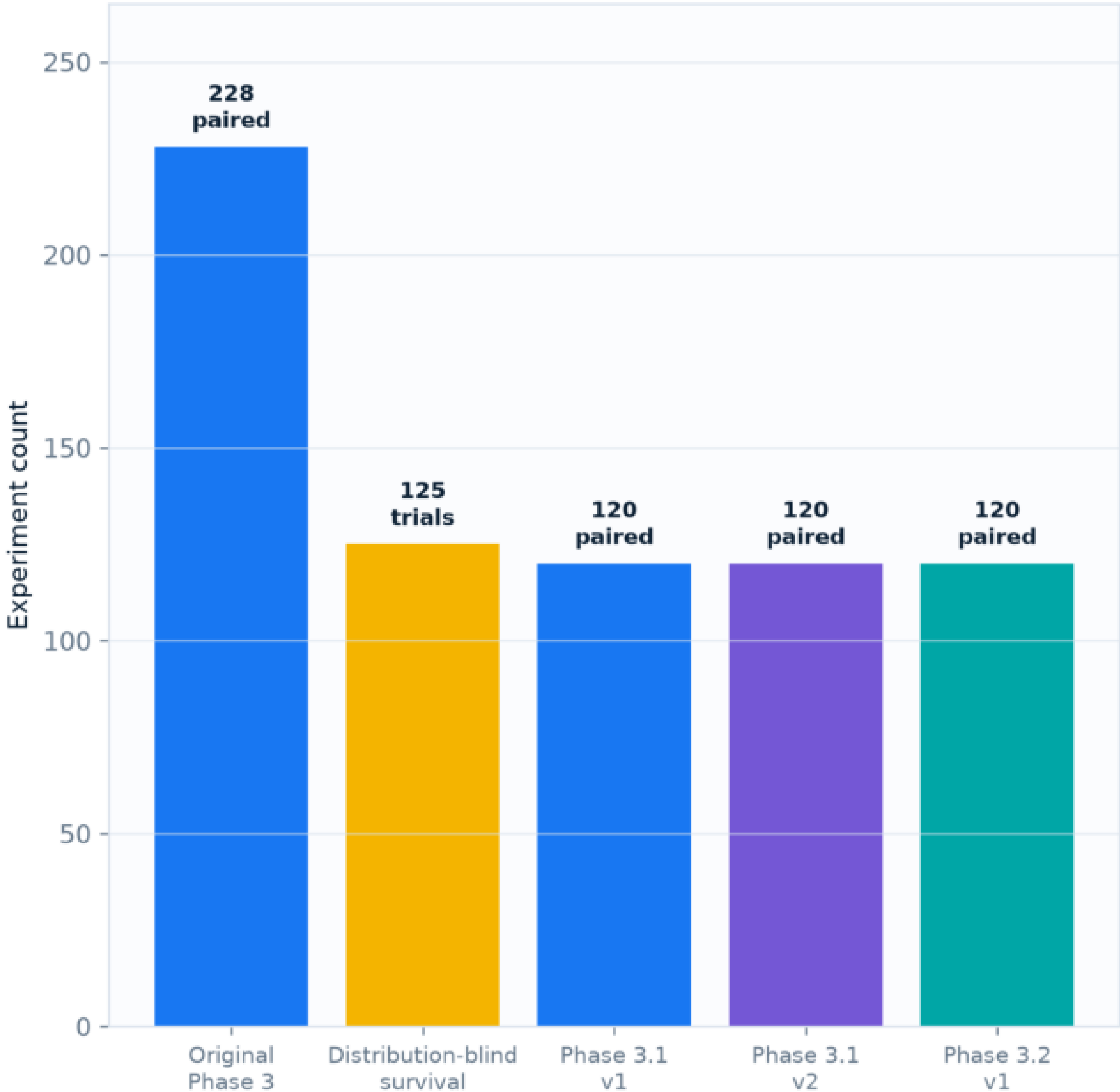


Fresh development seeds 46401–46472 · validation/release untouched

Campaign scale and release discipline

588 paired one-day controller runs + 125 distribution-blind survival trials · zero protected evaluations

All session experiments represented in this deck



Seed firewall held

Development consumed as registered

46101–46112 · 46401–46472

Validation UNTOUCHED

46201–46216

Release UNTOUCHED

46301–46330

Forecast test generated, never evaluated

46003

Fresh development seeds 46401–46472 · validation/release untouched

What worked—and what did not

Mechanism success is not promotion success · all claims remain bounded by measured evidence

WORKED	DID NOT WORK
✓ **Reproducibility** 29/29 frozen interfaces restored; content-addressed archives	✕ **Forecast promotion** Useful gains remained below the frozen 15% target
✓ **Lifecycle estimator** Observable export + Kaplan–Meier calibrated across 5 hidden distributions	✕ **MPC benefit** Exercised variants were neutral/slightly harmful
✓ **Fail-closed behavior** Stale survival returned to Static in 100% of trials	✕ **Generalization** Scheduled benefit did not transfer safely to mixed stress
✓ **Mechanism visibility** Proposed → certified → accepted → executed funnel captured	✕ **Tail safety** Strong blends missed the −10% worst-pair gate
✓ **Scheduled headroom** Pre-drain repeatedly reduced scheduled-fault overload	✕ **Adaptive protection** Surprise demand arrived after persistent-session commitment began
✓ **Release discipline** 588 paired runs, 28 candidates, zero unsafe promotions	✕ **Operational latency** Solver status hid end-to-end outliers up to 970 ms / MPC >6 s

FINAL DECISION · RETAIN STATIC · DO NOT CONSUME VALIDATION/RELEASE SEEDS

Fresh development seeds 46401–46472 · validation/release untouched

Final decision

Retain Static

What worked: reproducibility, lifecycle estimation mechanics, fail-closed controls, mechanism telemetry and release discipline.

What did not: robust cross-scenario benefit, tail safety, MPC material improvement and consistent operational latency.

Validation 46201–46216 · Release 46301–46330 · UNTOUCHED